

RESPIRATION

in PLANTS



Respiration is a **catabolic process** in which complex organic food molecules are broken down into simpler molecules with the **release of energy**.

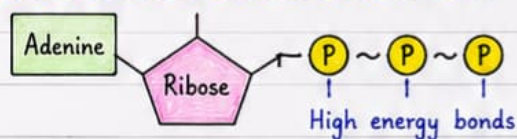
IMPORTANT POINTS

- Occurs in all living cells.
- Takes place **throughout the day**.
- Energy released is stored in the form of **ATP**.
- Both **aerobic** and **anaerobic** respiration occur in plants.
- Site of respiration - **Cytoplasm & Mitochondria**.

STEPS OF RESPIRATION

- 1 **Glycolysis (in Cytoplasm)**
Glucose (6C) → Pyruvic acid (3C)
2 ATP + NADH₂ produced.
- 2 **Link Reaction (in Mitochondrial matrix)**
• Pyruvic acid → Acetyl CoA
• CO₂ & NADH₂ released.
- 3 **Krebs Cycle (in Mitochondrial matrix)**
• Acetyl CoA completely oxidized.
• CO₂, NADH₂, FADH₂ & 2 ATP produced.
- 4 **Electron Transport Chain (on Inner Mitochondrial Membrane)**
• NADH₂ & FADH₂ donate e⁻.
• Large amount of ATP & H₂O formed.

ATP - ENERGY CURRENCY OF CELL

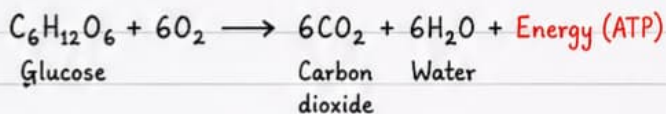


ATP stores energy in the bonds between phosphate groups. When a bond breaks, energy is released for cellular activities.

SIGNIFICANCE

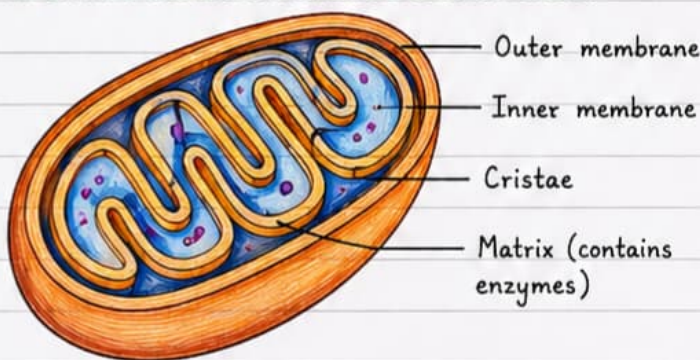
- Provides energy for all metabolic activities.
- Essential for **growth, reproduction & maintenance**.
- Helps in the synthesis of new molecules.

OVERALL EQUATION



During respiration, energy is released in a step-wise manner and is captured in the form of **ATP (Adenosine Triphosphate)**.

SITE OF RESPIRATION IN PLANT CELL



Mitochondria is the **powerhouse** of the cell where most of the respiration takes place.

TYPES OF RESPIRATION

- **Aerobic Respiration** - Requires O₂
(Complete breakdown of glucose)
$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$$
- **Anaerobic Respiration** - Does not require O₂
(Incomplete breakdown of glucose)
 - In plants (e.g., yeast): Ethanol + CO₂ + Energy
 - In some plant cells: Lactic acid + Energy

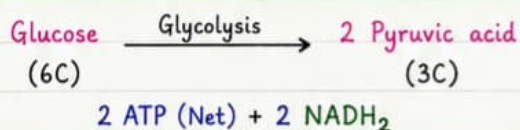
Examples

- Germinating seeds - high respiration rate.
- Root cells - absorb O₂ and respire.
- During heavy exercise in plants (e.g., flooding) - anaerobic respiration helps in survival.

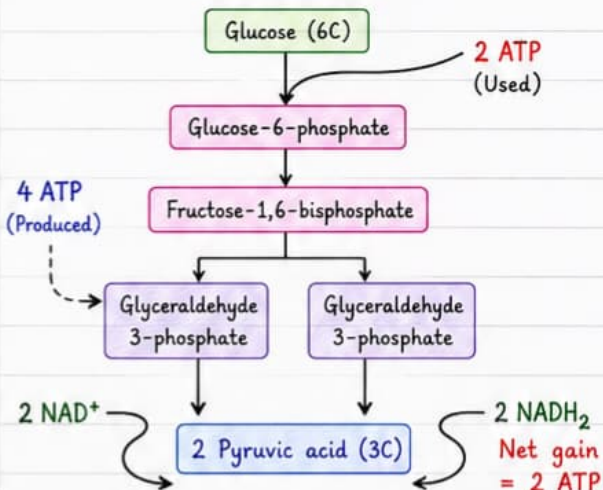


GLYCOLYSIS – FIRST STEP (IN CYTOPLASM)

- Glucose (6C) is activated using **2 ATP**.
- Glucose (6C) is split into two molecules of **Pyruvic acid (3C)**.
- During this process, **4 ATP** are produced but 2 are used → **Net gain = 2 ATP**.
- 2 NAD⁺** accept hydrogen → **2 NADH₂** are formed.



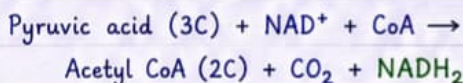
GLYCOLYSIS OVERVIEW



AFTER GLYCOLYSIS – IN MITOCHONDRION

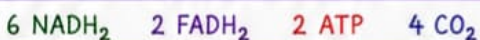
① Link Reaction (in Mitochondrial matrix)

- Pyruvic acid (3C) is converted into Acetyl CoA (2C).
- CO₂** is released.
- NAD⁺** is reduced to **NADH₂**.



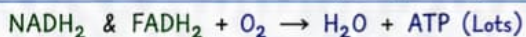
② Krebs Cycle / Citric Acid Cycle (in Matrix)

- Acetyl CoA enters the Krebs cycle.
- It gets completely oxidized to **CO₂**.
- NAD⁺**, **FAD** & **ADP + Pi** are converted to **NADH₂**, **FADH₂** & **ATP**.
- Per glucose, Krebs cycle produces:

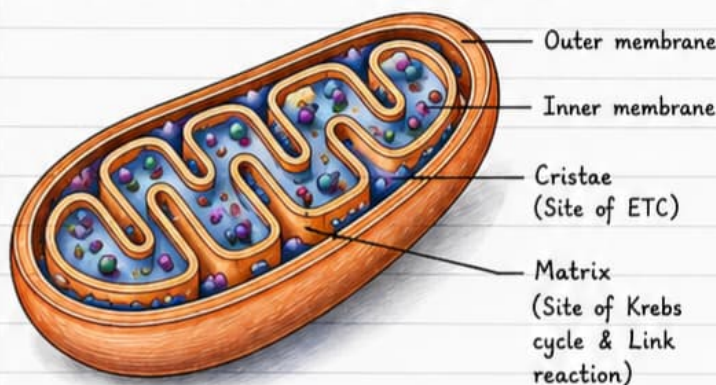


③ Electron Transport Chain (ETC) (on Inner Mitochondrial Membrane / Cristae)

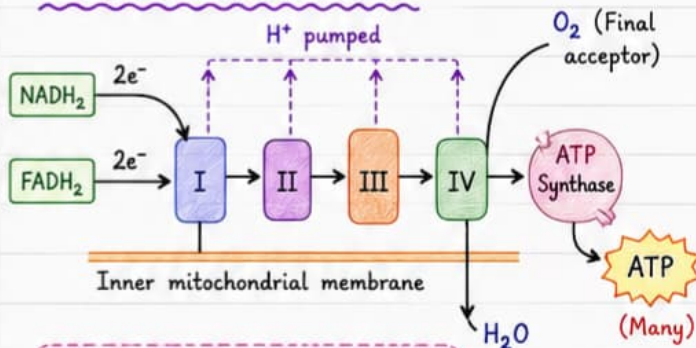
- NADH₂** & **FADH₂** donate electrons.
- Electrons move through carriers (ETC).
- Energy released is used to pump **H⁺** ions.
- H⁺** ions flow back through ATP synthase → Large amount of ATP is formed.
- O₂** acts as final electron acceptor.
- H⁺** combines with **e⁻** and **O₂** to form **H₂O**.



MITOCHONDRION – SITE OF RESPIRATION



ELECTRON TRANSPORT CHAIN



More ATP is produced in ETC as compared to earlier steps.

FACTORS AFFECTING RATE OF RESPIRATION

- | | |
|--|--|
| <ul style="list-style-type: none"> Temperature (optimum 25-35°C) Oxygen availability Water content Age of tissue | <ul style="list-style-type: none"> Type of tissue Light (indirectly) CO₂ concentration Inhibitors / Poisons |
|--|--|

ATP YIELD PER GLUCOSE MOLECULE

Glycolysis	2 ATP (Net)
Krebs Cycle	2 ATP
Electron Transport Chain	~ 34 ATP
TOTAL	~ 38 ATP

KEY TAKEAWAY

Respiration in plants is a catabolic process that breaks down glucose to release energy (ATP). It occurs in 4 major steps – **Glycolysis**, **Link Reaction**, **Krebs Cycle** & **Electron Transport Chain**.



ANAEROBIC RESPIRATION

- Occurs in **absence** of O_2 .
- Incomplete** breakdown of glucose.
- Takes place in the **cytoplasm**.
- Releases **less energy** (2 ATP per glucose).
- Seen in some plant cells (e.g., root tips in waterlogged soil), yeast & bacteria.

TYPES OF ANAEROBIC RESPIRATION IN PLANTS

(i) Alcoholic Fermentation

Occurs in yeast & some plant cells.

Glucose (6C)



2 Ethanol (2C) + 2 CO_2 + 2 ATP

(ii) Lactic Acid Fermentation

Occurs in some plant cells.

Glucose (6C)



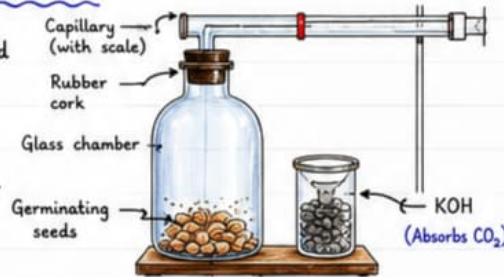
2 Lactic acid (3C) + 2 ATP

COMPARISON - AEROBIC vs ANAEROBIC RESPIRATION

Feature	Aerobic Respiration	Anaerobic Respiration
O_2 requirement	Required	Not required
Site	Mitochondria	Cytoplasm
Breakdown of glucose	Complete	Incomplete
End products	$CO_2 + H_2O$	Ethanol + CO_2 or Lactic acid
ATP produced per glucose	~ 36-38 ATP	2 ATP
Energy released	High	Low
Occurrence	Most plant cells (normal conditions)	Some plant cells (waterlogged conditions)

RESPIROMETRY - MEASURING RATE OF RESPIRATION

- The rate of respiration can be measured by the consumption of O_2 or evolution of CO_2 .
- Respirometer** is the instrument used.
- It helps in studying the effect of various factors on respiration.



Decrease in volume (rise of liquid in capillary) indicates O_2 consumption.

Working

- Seeds respire and consume O_2 .
- CO_2 produced is absorbed by KOH.
- Decrease in gas volume causes movement of liquid in capillary towards the chamber.
- Distance moved $\propto$ Rate of respiration.

Uses

- To measure rate of respiration.
- To compare respiration under different conditions.

FACTORS AFFECTING RATE OF RESPIRATION

1 Temperature :

Rate increases with rise in temperature up to optimum ($25-35^\circ C$), then decreases due to enzyme denaturation.



2 Oxygen availability :

More $O_2 \rightarrow$ higher rate of respiration.

3 Water content :

Moderate water is essential. Excess or deficiency reduces respiration.



4 Age of tissue :

Young, meristematic tissues respire faster than old tissues.

5 Concentration of CO_2 :

High CO_2 concentration may inhibit respiration.



6 Inhibitors / Poisons :

Compounds like cyanide, malonate etc. inhibit respiration by blocking enzymes.



ADAPTATIONS TO REDUCE RESPIRATION

- Storage organs (seeds, tubers, bulbs) are stored in cold and dry places.
- Fruits are stored at low temperature to slow down respiration and delay ripening.
- Reducing water content by drying or dehydration.
- Using chemicals (e.g., 2, 4-D in apples) to delay ripening.



SUMMARY

Respiration in plants is a catabolic process that releases energy in the form of ATP. It occurs in 4 steps - **Glycolysis** (cytoplasm) $\rightarrow$ Link Reaction, **Krebs Cycle** & **ETC** (mitochondria). Aerobic respiration produces ~ 36-38 ATP, while anaerobic respiration produces only 2 ATP. The rate of respiration is influenced by various internal and external factors.